Lightweight Requirements Capture Model Report

User Story + UseCase2.0 Combined

Abyss Navigator Suite Project

Triton Deepwater

01 April 2026

Document Control Item	Value
Report Type	Lightweight RCM report combining User Story and UseCase2.0 artifacts.
Progenitor	Team 1684R Project Description (PD) baseline for *The Abyss Navigator Suite*.
Customer	Triton Deepwater
User Story Convention	Connextra only: "As a . . . I want . . . so that . . . "
UC Story Convention	Alistair Cockburn casual / low-ceremony style
Priority Use Case	UC01 Handle Abnormal Conditions and Initiate Recovery
GPV Anchor	Safe autonomous deep-sea operations with controlled intervention, recovery readiness, and operational traceability.

Name	Student ID	Role Position
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Contents

Note on Hyperlinks: Throughout this document, internal references are displayed in blue and underlined. These links are fully clickable in the PDF viewer and navigate directly to the relevant section, table, diagram, or addendum.

1 Progenitor

This report is a lightweight derivative of the elicited Project Description (PD) for the *Abyss Navigator Suite*. The progenitor baseline establishes the project goal, scope, stakeholder context, Level 1 and Level 2 requirement statements, and the high-level models from which the stories, use case, and slices below were derived.

The applied progenitor baseline is summarized in Table 1.

Progenitor Aspect	Applied Baseline
Primary source	Team 1684R Project Description (PD) for *The Abyss Navigator Suite*.
Goal anchor	Enable unmanned deep-sea missions that are autonomous, safely supervised, recoverable, and auditable.
Scope anchor	Remote mission oversight, autonomous execution, safety handling, recovery coordination, and post-mission accountability.
Story source clauses	BO-L1-2, BO-L1-4, BO-L1-6, BO-L1-7, IT-L1-1, IT-L1-2, IT-L1-5, IT-L1-6 and related section text in the PD.
Method selection	Connextra for User Stories; Cockburn casual/low-ceremony for UC Stories; STS-style slices for realization planning.

Table 1: Applied PD progenitor baseline for this lightweight report

2 Executive Summary

This report applies a deliberately lightweight RCM approach to the Abyss Navigator Suite by combining small, business-facing User Stories with one system-wide Use Case diagram and one deeper UseCase2.0 drill-down. The User Story portion captures concise operational needs that still carry business value: mission configuration, abnormal-condition alerting, and post-mission accountability. Each story stays traceable to the progenitor PD so the report remains concise without losing chain-of-custody back to the original elicited need.

The UseCase2.0 portion keeps the big picture through one GPV-aligned diagram that relates four actors, namely the Mothership Pilot, Emergency Retrieval System, Fleet Operations Controller, and Mission Review Officer, to seven customer-visible goals: UC01 Handle Abnormal Conditions and Initiate Recovery, UC02 Monitor Vessel Health, UC03 Manual Control Override, UC04 Execute Mission Autonomously, UC05 Configure Mission Parameters, UC06 Manage Fleet Mission, and UC07 Review Mission Record. The report then drills into **UC01 Handle Abnormal Conditions and Initiate Recovery** because it is the most safety-critical use case in the current model. Within that priority use case, the walking skeleton is the thinnest monitor-to-alert backbone that proves the safety-response path is technically alive, while the MVP is the first usable response flow that couples alerting, override routing, and recovery handoff. The resulting UC stories, auxiliary detail, conversation notes, and two STS-style slices keep the document aligned to the lecture approach: one lightweight whole-system view, then one priority use case elaborated just deeply enough for realization planning.

3 User Story RCM

Chosen convention. This report uses the **Connextra** convention only. Every User Story follows the exact pattern *As a . . . I want . . . so that . . . .* No hybridization with INVEST or Three-C's is used in the story wording itself.

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The story set, its PD references, and its business confirmation signals are consolidated in Table 2.

ID	Connextra User Story	PD Reference	Business Confirmation Signal
US-01	As a fleet operations controller, I want to configure mission parameters before deployment so that each autonomous mission starts with approved objectives, route settings, and operational limits.	PD 1.5 Project Goal; PD 2.1 In Scope; PD 6.1 Autonomous Mission Execution; PD 7.4 IT Capability Requirements.	The controller can create and save a mission configuration that is ready for autonomous execution.
US-02	As a mothership pilot, I want critical pressure, temperature, and power deviations to raise a clear health alert so that I can initiate recovery before risk escalates.	PD 1.5 Project Goal; PD 2.1 In Scope; PD 4.3 Regulatory Obligations; PD 6.4 Emergency and Abnormal Situation Handling; BO-L2-4.1; BO-L2-4.2.	A critical health alert appears with vehicle ID, mission ID, breached parameter, severity, and timestamp.
US-03	As a mission review officer, I want to review a completed mission record with anomalies and overrides so that I can conduct post-mission review and investigation.	PD 2.1 In Scope; PD 6.6 Operational Accountability; PD 6.7 Mission Data Value Delivery; PD 7.2 Data Management; IT-L1-5; IT-L2-5.1.	The review officer can open a completed mission and see an attributable timeline of mission events, anomalies, and override actions.

Table 2: Three lightweight User Stories derived from the PD progenitor baseline

In this report, “PD” refers to the team’s Project Description and its progenitor baseline.

Full progenitor trace links for the story set are consolidated in Table 8 in Addendum A: Progenitor Trace Links.

User Story Conversation: US-02

Before implementation of US-02, the following stakeholder clarifications remain necessary:

- Exact threshold ownership: who approves the operational baseline for pressure, temperature, and power thresholds when sea-trial data changes?
- Alert semantics: should repeated breaches for the same parameter open a new alert each time or update an existing active alert?
- Escalation rule: when does the system only raise a health alert, and when must it recommend immediate recovery initiation?
- Human authority: which roles may acknowledge the alert, request manual override, or initiate recovery during a live mission?
- Evidence rule: what minimum incident detail must be retained for later compliance review and investigation?

4 UseCase2.0 RCM

Whole-System Use Case Diagram

The whole-system use case diagram is provided in Addendum C. In the current model, the Mothership Pilot and Emergency Retrieval System participate in the safety-response cluster around UC01, UC02, and UC03; the Fleet Operations Controller owns mission execution and planning use cases UC04, UC05, and UC06; and the

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Mission Review Officer owns UC07 Review Mission Record. This keeps the diagram aligned to the progenitor GPV while staying within the seven-oval limit required by the rubric.

Within the current model, UC01 Handle Abnormal Conditions and Initiate Recovery is the only use case selected for detailed treatment in this report because it connects the monitoring, override, and recovery concerns into one business-critical safety response. The remaining use cases shown in Addendum C: UC02 Monitor Vessel Health, UC03 Manual Control Override, UC04 Execute Mission Autonomously, UC05 Configure Mission Parameters, UC06 Manage Fleet Mission, and UC07 Review Mission Record, are retained at whole-system level and are not decomposed further in this document.

Priority Use Case, Walking Skeleton, and MVP

The priority use case, walking skeleton, and MVP definitions are summarized in Table 3.

Item	Definition
Priority use case	**UC01 Handle Abnormal Conditions and Initiate Recovery.** In the current diagram, this use case best concentrates the project's GPV because it turns monitoring outcomes into an accountable safety response that can either trigger recovery or route the incident into control intervention. See Addendum C and the UC01 stories.
Walking skeleton	**UC-01-WS: Monitor-to-alert backbone.** The thinnest end-to-end proof that a critical health event from UC02 Monitor Vessel Health can trigger UC01, raise a visible alert, and record the incident. This item is a prioritization construct rather than a story and is realized in UC Slice 1.
MVP	**UC-01-MVP: First usable safety response.** The smallest version Triton Deepwater can meaningfully use: accept critical monitoring events, display a health alert, route the incident into recovery initiation or UC03 Manual Control Override, update mission status, notify the Emergency Retrieval System when recovery is chosen, and retain incident context. This item is a prioritization construct rather than a story and is realized in UC Slice 2.

Table 3: Priority use case definition and prioritization scheme

Further elaboration of UC01 is provided in the UC stories section, the auxiliary detail table (Table 4), the UC conversation section, the walking skeleton slice table (Table 5), and the MVP slice table (Table 6).

UC Stories for UC01 Handle Abnormal Conditions and Initiate Recovery

UC01-S1 Basic Story: A critical condition leads to recovery initiation. During autonomous mission execution, UC02 Monitor Vessel Health detects a parameter that exceeds the approved threshold. UC01 takes ownership of the incident, raises a health alert to the mothership pilot, and coordinates the first recovery response with the emergency retrieval system.

Main scenario

1. The mission is active under autonomous execution and vessel-health monitoring is live for the selected submersible.
2. A pressure, temperature, or power value breaches the configured critical threshold.
3. UC01 records the abnormal condition and opens a health alert for the affected mission and vehicle.
4. The system presents the alert to the mothership pilot with severity, breached parameter, timestamp, and recommended response.
5. The mothership pilot confirms recovery initiation.

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6. The system changes mission state to recovery initiated and sends the recovery handoff to the emergency retrieval system.
7. The system records the handoff outcome and keeps the incident visible until recovery status is confirmed.

UC01-S2 Alternate Story: The incident is routed into UC03 Manual Control Override before full recovery. The abnormal condition is visible, but the operator decides the vehicle should be stabilized or repositioned under manual control before committing to recovery. The system records the override decision, suspends purely autonomous control, routes the incident into the manual-control path, and keeps the abnormal condition open until recovery is later initiated or the health state returns to an acceptable range.

UC01-S3 Exception Story: Vessel-health data is unavailable or contradictory. UC02 cannot produce a trustworthy health state because the incoming signal is missing, stale, or malformed. The system raises a sensor-data fault, marks the health state as uncertain, and prevents the abnormal event from being treated as resolved without human review.

UC Auxiliary Detail for UC01

The auxiliary detail set for UC01 is consolidated in Table 4.

Auxiliary Detail Item	Content
For / effective date	Use Case **UC01 Handle Abnormal Conditions and Initiate Recovery**; effective date **2026-091**.
Assumptions	Missions may already be underway under UC04 Execute Mission Autonomously and are supervised from Triton Deepwater's surface or mothership environment. Triton-provided sensors remain available for software integration through UC02 Monitor Vessel Health. A configured baseline exists for each monitored parameter before mission launch. A recovery handoff path to the emergency retrieval system and an authorized operator session are available.
Business rules	Critical abnormal conditions must be escalated through UC01 and not remain hidden inside monitoring alone. Every alert, override transition, and recovery-initiation action must remain attributable to mission, vehicle, time, and actor. Routing an incident into UC03 Manual Control Override does not close the incident; it only changes the response mode until the condition is resolved or recovery is initiated.
Constraints	Communications may degrade to low bandwidth, intermittent connectivity, and up to 30-second one-way latency. Deep-sea missions remain unmanned. Compliance with maritime safety and data-governance obligations is mandatory.
Pre- & post-conditions	**Pre:** mission run exists, threshold profile is loaded, UC02 vessel-health monitoring is active, and a pilot session plus retrieval-system channel are available. **Post:** abnormal event is persisted; one of recovery initiated, manual override active, or sensor-fault escalated is visible; mission state is updated to reflect the chosen response; and audit history is complete enough for investigation.
Technical details: database elements & parameters	MissionRun, TelemetrySnapshot, ThresholdProfile, HealthAlert, OverrideCommand, RecoveryEvent, and OperationalLog. Pressure deviation $> \pm 5\%$ within 5 seconds is critical. Temperature deviation $> \pm 10\%$ within 10 seconds is critical. Power deviation $> \pm 15\%$ within 10 seconds is critical.
Technical details: load & stress parameters	Support at least 3 supervised submersibles in one session. Continue operating under constrained communication at 1 kbps minimum bandwidth, 30-second one-way latency, and intermittent link conditions.

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Auxiliary Detail Item	Content
Technical details: performance benchmarks	Telemetry refresh target is every 30 seconds when connectivity is available. Pressure anomaly detection target is within 5 seconds; temperature and power within 10 seconds. Recovery-initiation or override acknowledgement target is within 60 seconds when link latency is at or below 30 seconds.
Technical details: platform & IT compatibility	Telemetry and sensor integration use ROS2 middleware, DDS transport, and JSON schema validation. The operator interface runs from Triton Deepwater's surface or mothership environment. Operational records are stored in PostgreSQL or equivalent structured storage.
Technical details: quality features	Safety integrity, traceability, operability, resilience under degraded communications, and confidentiality/integrity of operational data.
Technical details: UI & UX	The incident panel must show severity, vehicle ID, mission ID, breached parameter, timestamp, and recommended next action on one screen. The manual-override and initiate-recovery controls must remain visible while the incident is open, and the operator must be able to tell whether the current response mode is autonomous, manual override, or recovery initiated.

Table 4: Auxiliary detail list for the priority use case

Quantitative source notes that support these auxiliary details are consolidated in Table 9 in Addendum B: Quantitative and Interface Source Notes.

UC Conversation

Before realization slices are finalized, the following matters require clarification from Triton Deepwater stakeholders:

- Whether the mothership pilot alone can confirm recovery initiation, or whether the emergency retrieval system must acknowledge the trigger before UC01 is considered complete.
- Whether routing an incident into UC03 Manual Control Override automatically suspends UC04 autonomous execution or requires a second confirmation step.
- Which vessel-health indicators from UC02 must remain visible during a live abnormal-condition incident and subsequent recovery handoff.
- The exact rule for merging, reopening, or auto-closing repeat health alerts on the same parameter.
- Whether a sensor-data fault should immediately pause autonomous execution or remain as a warning pending human review.

UC Slice 1: Walking Skeleton

Title: UC01-SL01 Monitor-to-Alert Backbone

The walking skeleton slice definition is detailed in Table 5.

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STS Field	Slice Content
Context	**Effective date:** 2026-091. **Parent UC:** UC01 Handle Abnormal Conditions and Initiate Recovery. **Story list:** UC01-S1. **Prelim:** Seed one active autonomous mission, one vehicle, one threshold profile, one pilot session, and one UC02 monitoring input path. Use a telemetry simulator or stub that can emit one pressure breach event.
Narrative	The slice proves the first safety-response backbone is alive. A monitoring event from UC02 enters through the boundary, passes through abnormal-condition evaluation logic in UC01, creates one health-alert entity, appears on the pilot incident panel, and records one alert event in the audit trail.
Scope	**Category:** Basic. **Priority:** Blocker. **Status:** Walking Skeleton. **KIV:** recovery dispatch, manual override transition, multi-parameter rules, and degraded-link handling beyond basic alert visibility.
Tech details: Boundary	UC02 monitoring input stub for one vehicle; mission health or incident widget showing the current alert to the mothership pilot.
Tech details: Control	MonitorVesselHealthAdapter validates incoming payload; AbnormalConditionService compares against threshold profile; HealthAlertController exposes the incident state; AuditService records the alert event.
Tech details: Entity	MissionRun, TelemetrySnapshot, ThresholdProfile, HealthAlert, and OperationalLog tables or equivalents.
Tech details: Interfaces / infrastructure	ROS2 or simulator adapter into the application boundary; dashboard update over internal API; local PostgreSQL schema and seeded mission data; baseline health check for the service stack.
Acceptance	AC01 [Y/N]: One simulated pressure breach creates exactly one health alert linked to the correct mission and vehicle. AC02 [Y/N]: The alert appears on the dashboard within 5 seconds of ingest. AC03 [Y/N]: The alert event is written to the audit log with timestamp and vehicle context. AC04 [Y/N]: The end-to-end path boundary → control → entity is demonstrably online.

Table 5: STS-style walking skeleton slice for the priority use case

UC Slice 2: MVP

Title: UC01-SL02 First Usable Safety Response

The MVP slice definition is detailed in Table 6.

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STS Field	Slice Content
Context	**Effective date:** 2026-091. **Parent UC:** UC01 Handle Abnormal Conditions and Initiate Recovery. **Story list:** UC01-S1, UC01-S2, UC01-S3. **Prelim:** Walking skeleton slice is working; threshold profiles for pressure, temperature, and power are configured; command adapters support recovery initiation and manual override; UC04 mission execution is already running; mission UI already shows active health alerts.
Narrative	This slice delivers the first usable safety-response flow. Critical monitoring events from UC02 open a health alert in UC01, the operator can route the incident either to recovery initiation or into UC03 Manual Control Override, mission state is updated, the emergency retrieval system receives the recovery handoff when selected, and the incident history is preserved for later review.
Scope	**Category:** Basic. **Priority:** High. **Status:** MVP. **KIV:** automated emergency retrieval orchestration, fleet-wide alarm triage, richer sensor-fault diagnostics, and downstream regulatory reporting automation.
Tech details: Boundary	Mission health panel with live alert list, recommended-action text, manual-override control, initiate-recovery control, degraded-link indicator, and recovery status card.
Tech details: Control	Multi-parameter rule evaluation for pressure/temperature/power; role-based authorization check for operator actions; AutonomousExecutionStateManager for suspending autonomous mode when override is chosen; ManualOverrideService for UC03 transition; RecoveryDispatchService for recovery request; ConnectivityMonitor for degraded-link state; RecoveryStatusService for last-known position and handoff visibility.
Tech details: Entity	MissionRun, ThresholdProfile, HealthAlert, OverrideCommand, RecoveryEvent, OperationalLog, and TelemetrySnapshot.
Tech details: Interfaces / infrastructure	Command channel to the emergency retrieval system and override adapter; cached read model for last-known mission state; JSON schema validation on inbound telemetry; database migration for alert, override, and recovery entities; environment configuration for latency and outage simulation.
Acceptance	AC11 [Y/N]: Pressure, temperature, and power breaches produce the correct health-alert severity and recommended action. AC12 [Y/N]: The mothership pilot can initiate recovery from an active health alert and the emergency retrieval system receives notification within 60 seconds under simulated 30-second latency. AC13 [Y/N]: If manual override is chosen, mission state changes to `MANUAL_OVERRIDE_ACTIVE`, autonomous execution is suspended or marked paused, and the incident remains open. AC14 [Y/N]: If the link degrades after alert creation, the dashboard preserves the last-known incident context and logs the degraded communication event.

Table 6: STS-style MVP slice for the priority use case

5 Combined Sign-off

This Lightweight RCM Report is certified as complete and ready for submission to Triton Deepwater.

Signed by:

Muhammad Ridwan Putra Jasni
Project Leader
01 April 2026

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6 Declarations

AI Use Declaration: ChatGPT was used for language support, grammar refinement, and explanation of complex concepts. All stories, use cases, slices, models, and diagrams were produced by the team.

Originality Declaration: This report presents original lightweight requirements capture work for the Abyss Navigator Suite project and is derived from the team's PD baseline.

Team declarations and signatures are recorded in Table 7.

Name	Student ID	Signature
Muhammad Ridwan Putra Jasni	2401684	
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Table 7: Team declarations and signatures

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A Addenda

A.1 Addendum A: Progenitor Trace Links

This addendum supports the base User Story and UseCase2.0 sections by preserving concise trace links back to the PD without expanding the body into a heavyweight trace matrix.

Item	Base Section	PD / Progenitor Link	Why It Matters
US-01	User Story RCM	PD 1.5; PD 2.1; PD 6.1; PD 7.4; IT-L1-1.	Preserves mission setup as an explicit controller-facing business need before autonomous execution begins.
US-02	User Story RCM	PD 1.5; PD 2.1; PD 4.3; PD 6.4; BO-L2-4.1; BO-L2-4.2.	Keeps the alerting story tied directly to safety and compliance intent.
US-03	User Story RCM	PD 2.1; PD 6.6; PD 6.7; PD 7.2; IT-L1-5; IT-L2-5.1.	Preserves accountability and post-mission value as explicit business needs.
UC-01	UseCase2.0 RCM	PD 1.5; PD 2.1; PD 6.4; PD 6.5; PD 7.4.	Justifies selecting abnormal-condition handling and recovery initiation as the priority use case in the current diagram.
UC01-WS	UC Slice 1	BO-L2-4.1 plus IT-L1-1 / IT-L1-5 support.	Confirms the first slice proves the monitor-to-alert backbone before override or recovery-handoff behavior.
UC01-MVP	UC Slice 2	BO-L2-4.1, BO-L2-4.2, IT-L1-2, IT-L1-6.	Shows why the MVP must integrate monitoring, override routing, and recovery handoff under real link constraints.

Table 8: Trace links from lightweight artifacts back to the PD progenitor baseline

A.2 Addendum B: Quantitative and Interface Source Notes

This addendum supports the base UC Auxiliary Detail and UC Slice sections by holding the quantitative notes and interface assumptions used to keep the priority use case CRaM and testable.

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Note ID	Base Section	Supporting Note	Source
ADD-B1	UC Auxiliary Detail	Pressure deviations beyond $\pm 5\%$ are treated as critical within 5 seconds.	PD 1.5; PD 6.4; BO-L2-4.1.
ADD-B2	UC Auxiliary Detail	Temperature deviations beyond $\pm 10\%$ and power deviations beyond $\pm 15\%$ are treated as critical within 10 seconds.	PD 1.5; PD 6.4; BO-L2-4.2.
ADD-B3	UC Auxiliary Detail / Slice 2	Telemetry should refresh at least every 30 seconds when connectivity is available.	PD 2.1; PD 7.4; IT-L1-1.
ADD-B4	UC Auxiliary Detail / Slice 2	Recovery-initiation or manual-override acknowledgement target is within 60 seconds when one-way latency is at or below 30 seconds.	PD 6.4; PD 7.4; IT-L1-2; IT-L1-6.
ADD-B5	UC Auxiliary Detail / Slice 2	Degraded communications planning constrains alert, override, and recovery handling under low bandwidth, high latency, and sustained link loss.	PD 2.1; PD 7.4; IT-L1-6.
ADD-B6	UC Auxiliary Detail / Slice 1	Telemetry integration assumes ROS2 middleware, DDS transport, and JSON schema validation.	PD 7.4; IT-L1-1; IT-L1-2.

Table 9: Quantitative and interface notes supporting the priority use case

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A.3 Addendum C: Whole-System Use Case Diagram

s This addendum contains the GPV-aligned seven-use-case diagram referenced in the UseCase2.0 section. It relates the Mothership Pilot, Emergency Retrieval System, Fleet Operations Controller, and Mission Review Officer to UC01 through UC07.

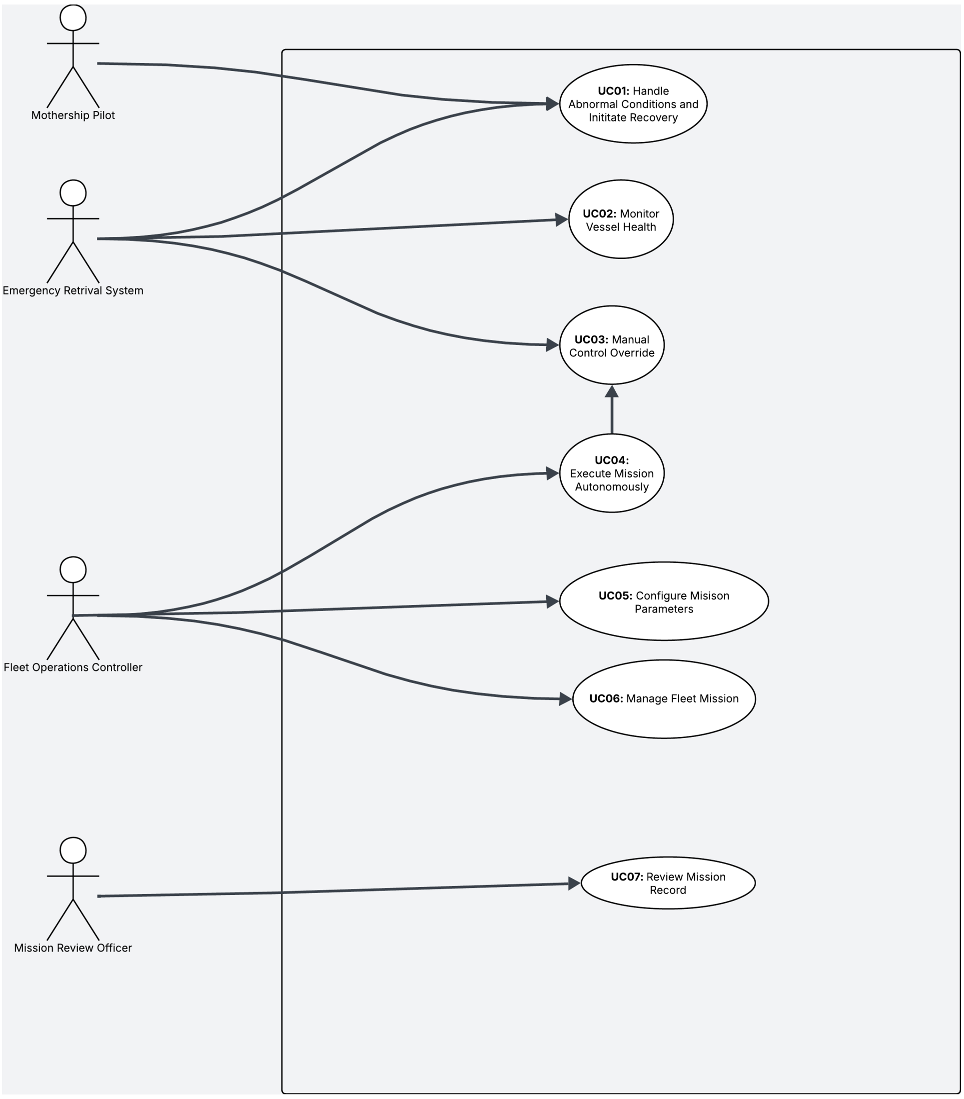

Figure 1: GPV-aligned whole-system use case diagram used by the lightweight UseCase2.0 section

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